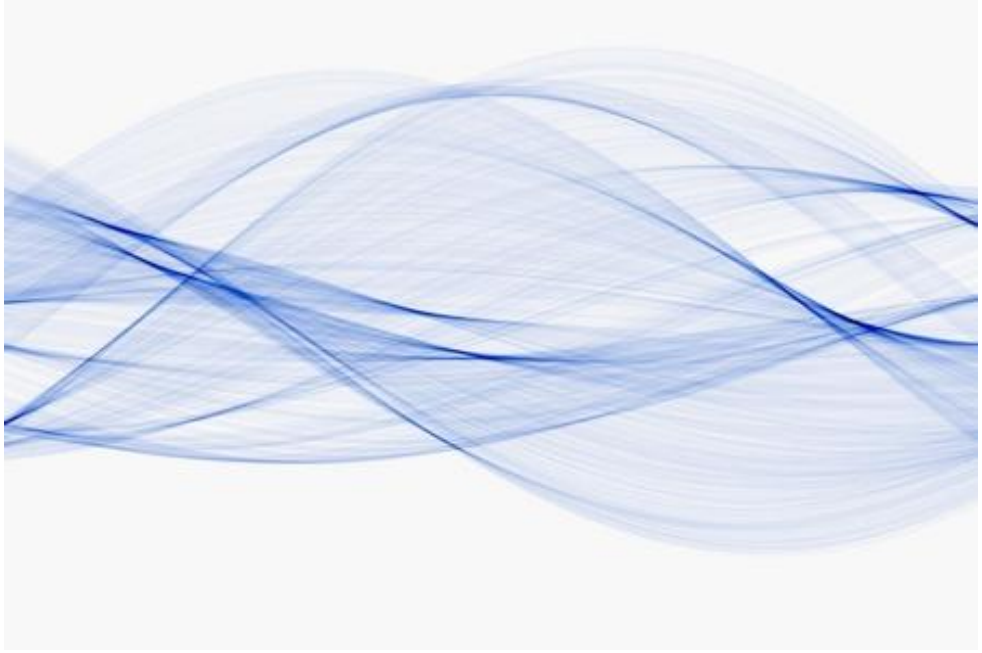
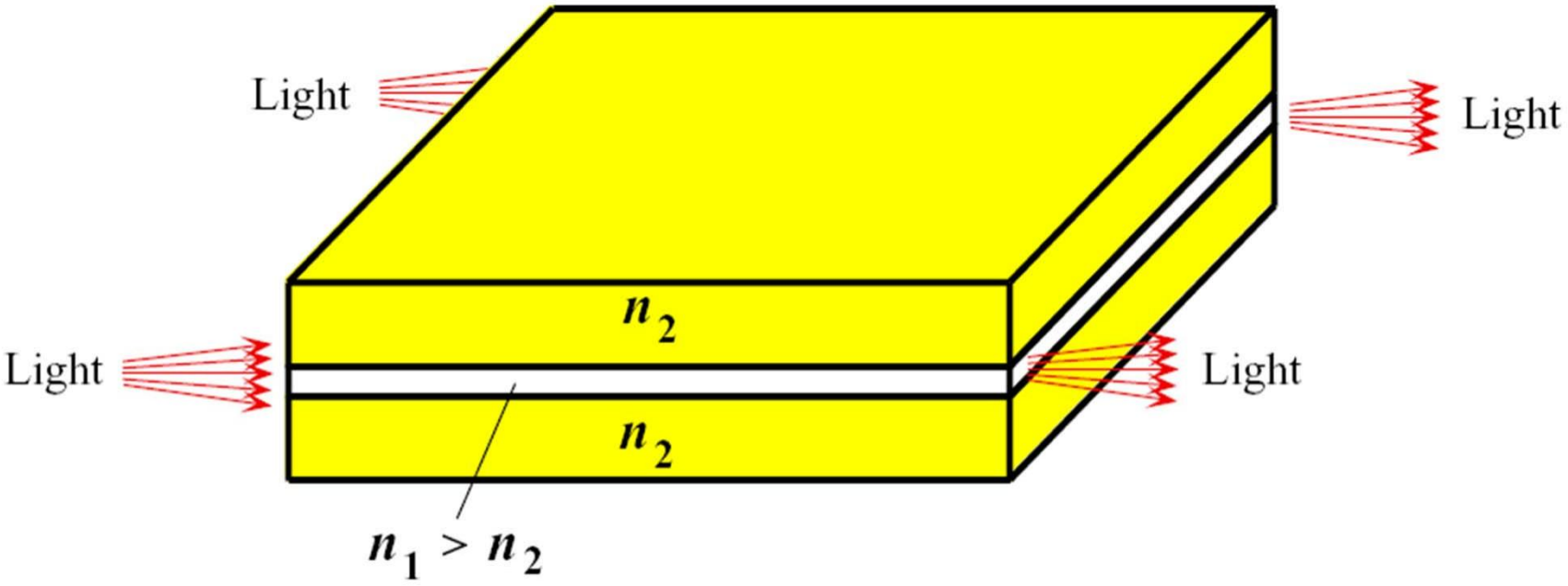
Dielectric Waveguides and Optical Fiber

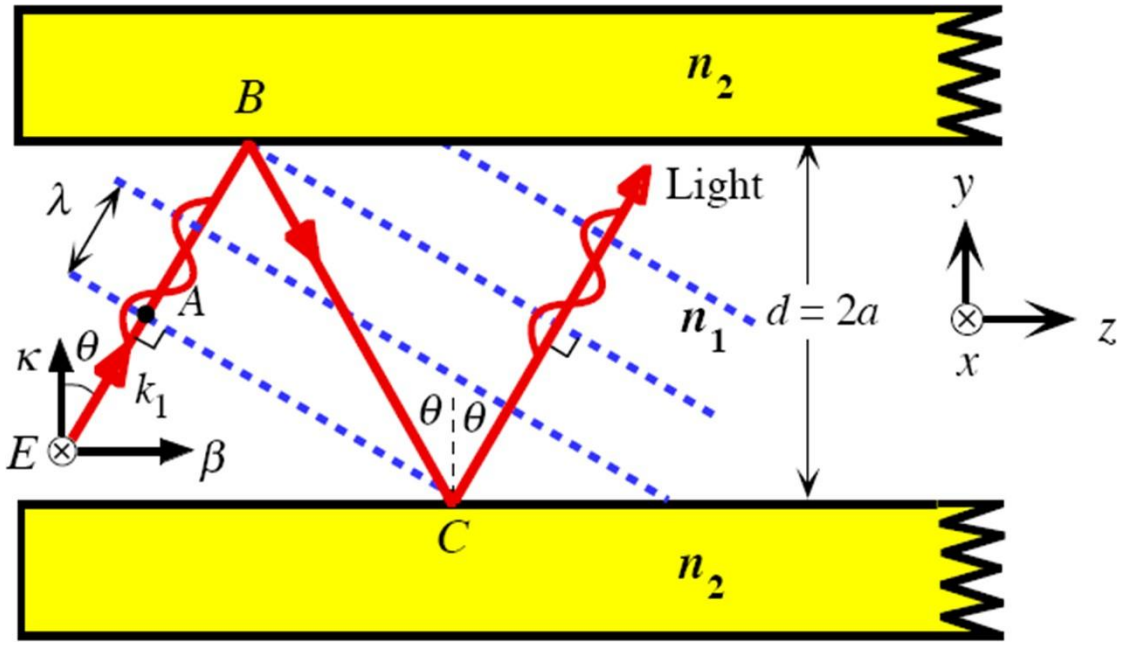
Dr. Monir Morshed
Professor
Dept. of ICT, MBSTU



Symmetric Planar Dielectric Slab Waveguides



- The region of a higher refractive index (n_1) is called the core,
- The region of a lower refractive index (n_2) sandwiching the core is called the cladding.
- For light rays to propagate within the core, total internal reflection (TIR) should occur. Our goal is to find the conditions for light rays to propagate along such waveguides.



A and C are on the same wavefront. They must be in phase.

OR

Constructive interference occurs between A and C to achieve maximal transmitted light intensity.

$$\Delta\phi(\mathrm{AC}) = k_1(\mathrm{AB}+\mathrm{BC}) - 2\phi$$

$$= k_1[\mathrm{BC}\cos(2\theta)+\mathrm{BC}] - 2\phi$$

$$= k_1 \frac{d}{\cos\theta}\left[2\cos^2\theta - 1 + 1\right] - 2\phi$$

$$= k_1[2d\cos\theta] - 2\phi$$

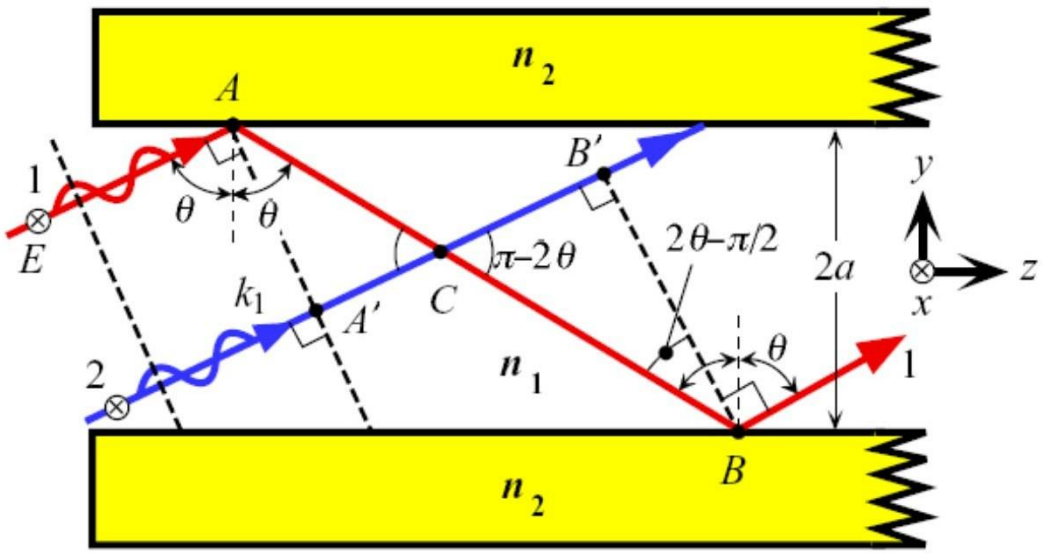
$$\Rightarrow k_1[2d\cos\theta] - 2\phi = m(2\pi), m = 0,1,2,\ldots$$

$$\Delta\phi(\mathrm{AC}) = m(2\pi),\ m=0,1,2,\ldots$$

Waveguide condition:

$$\left\lfloor \frac{2\pi n_1(2a)}{\lambda} \right\rfloor \cos\theta_m - \phi_m = m\pi$$

ϕ_m is a function of θ_m.



B and B' are on the same wavefront. They must be in phase.

OR

Constructive interference occurs between B and B' to achieve maximal transmitted light intensity.

$$\Delta\phi = k_1(\mathrm{AB}) - 2\phi - k_1(\mathrm{A'B'})$$

$$= k_1(\mathrm{AB}) - k_1[\mathrm{AB}\cos(\pi - 2\theta)] - 2\phi$$

$$= k_1(\mathrm{AB})(2\cos^2\theta) - 2\phi$$

$$= k_1 \frac{d}{\cos\theta}[2\cos^2\theta] - 2\phi$$

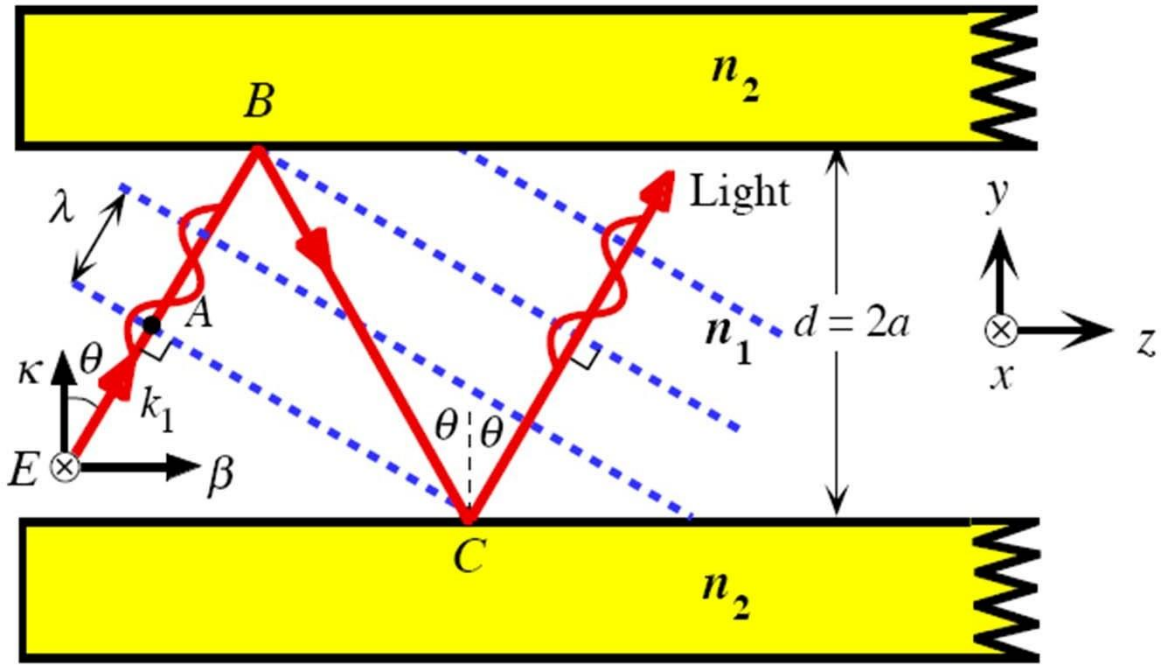
$$\Rightarrow k_1[2d\cos\theta] - 2\phi = m(2\pi), m = 0,1,2,...$$

$\Delta\phi(\mathrm{BB'}) = m(2\pi)$, m=0,1,2,…

Waveguide condition:

$$\left\lfloor \frac{2\pi n_1(2a)}{\lambda} \right\rfloor \cos\theta_m - \phi_m = m\pi$$

Propagation Constant along theWaveguide



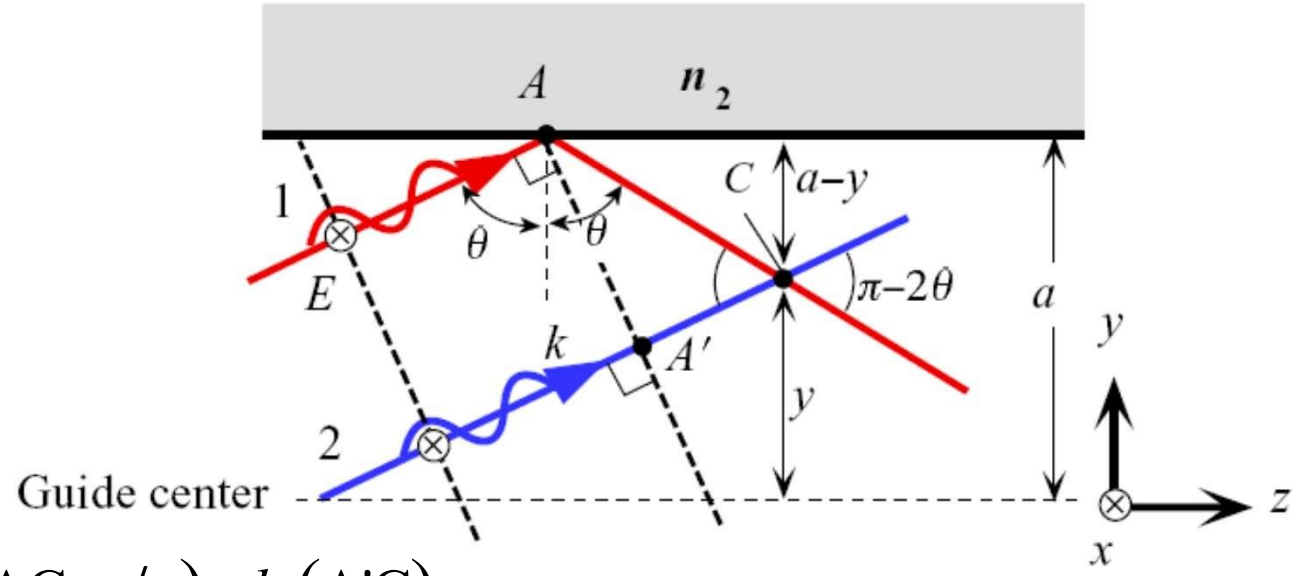
$$\begin{cases} \beta_m = k_1 \sin\theta_m = \left(\dfrac{2\pi n_1}{\lambda}\right)\sin\theta_m \\ \kappa_m = k_1 \cos\theta_m = \left(\dfrac{2\pi n_1}{\lambda}\right)\cos\theta_m \end{cases}$$

κ_m: transverse propagation constant

$$\left[\frac{2\pi n_1 (2a)}{\lambda}\right]\cos\theta_m - \phi_m = m\pi$$

Larger m leads to smaller θ_m.

Electric Field Patterns



$$\Phi_m = (k_1 \mathrm{AC} - \phi_m) - k_1(\mathrm{A'C})$$

$$= k_1 \mathrm{AC} - k_1 \mathrm{AC}\cos(\pi - 2\theta_m) - \phi_m$$

$$= k_1 \mathrm{AC}\left(2\cos^2\theta_m\right) - \phi_m$$

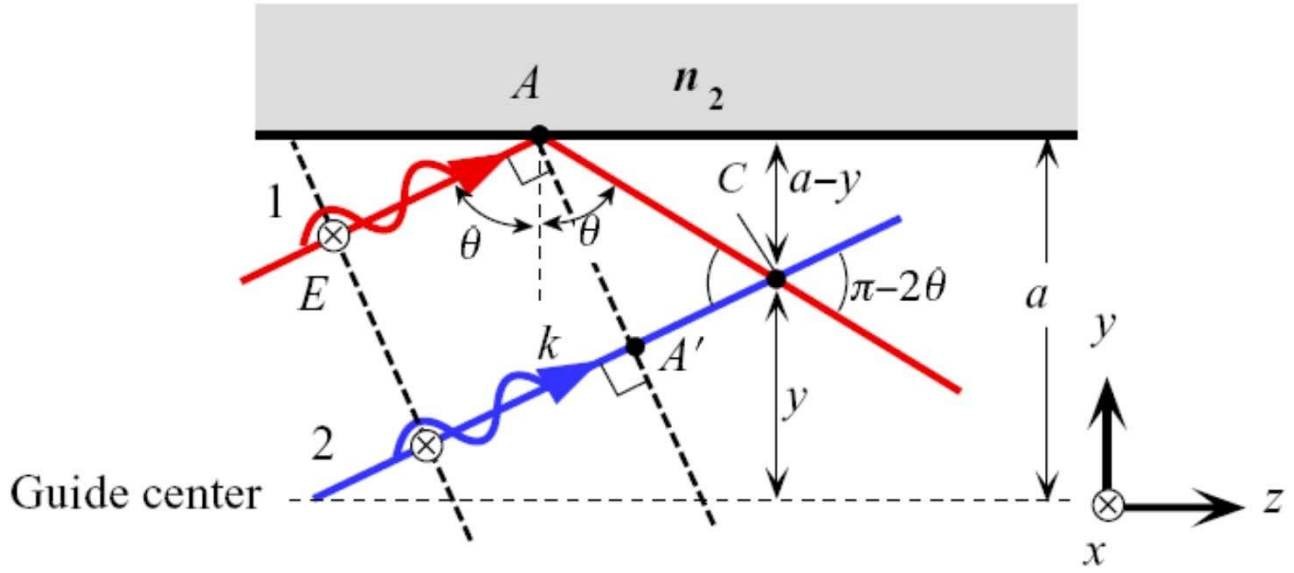
$$= k_1 \frac{a-y}{\cos\theta_m}\left[2\cos^2\theta_m\right] - \phi_m$$

$$= 2k_1(a-y)\cos\theta_m - \phi_m$$

$$\left\lfloor \frac{2\pi n_1 (2a)}{\lambda} \right\rfloor \cos\theta_m - \phi_m = m\pi$$

$$\Phi_m(y) = m\pi - \frac{y}{a}(m\pi + \phi_m)$$

Electric Field Patterns



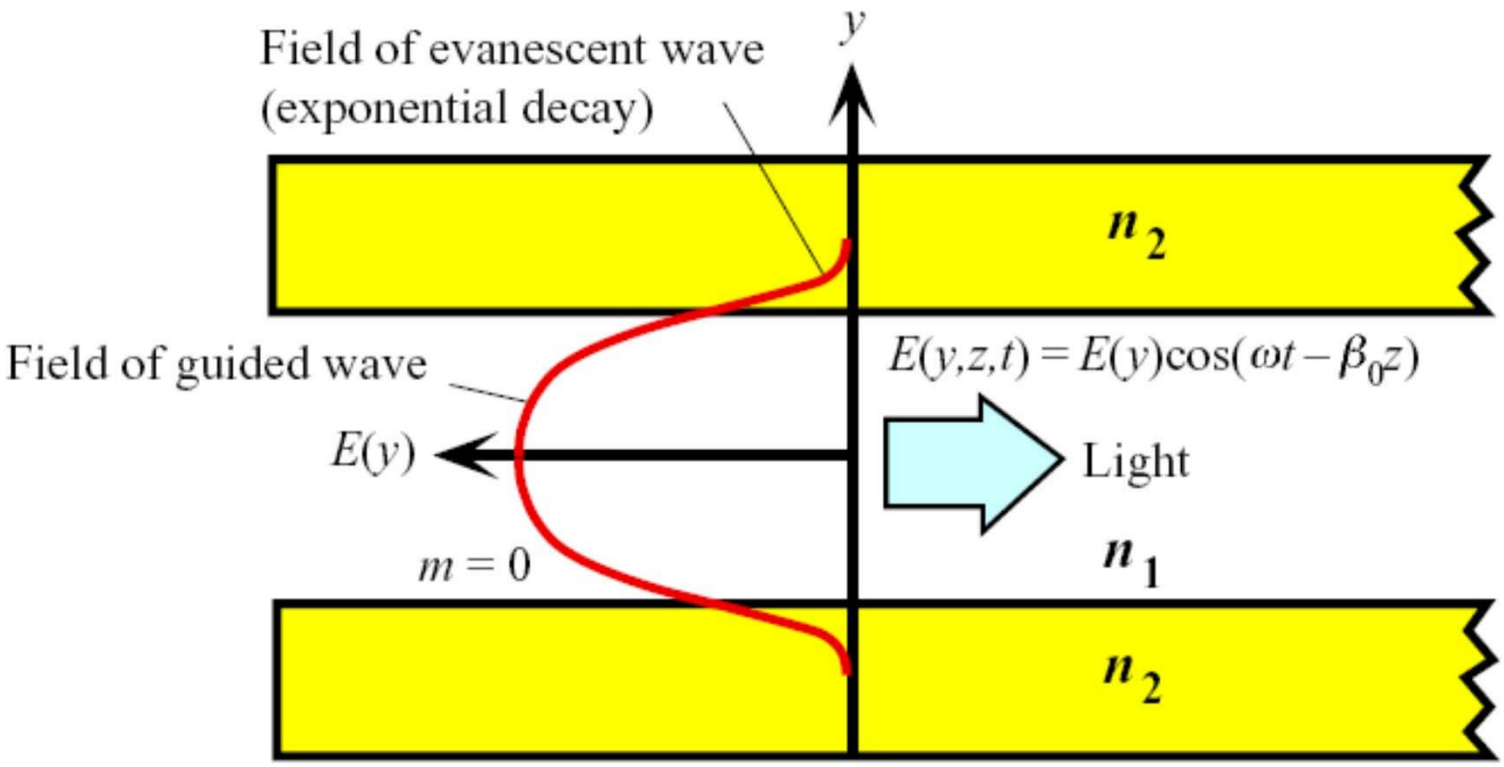
$$E_1(y,z,t) = E_0 \cos(\omega t - \beta_m z + \kappa_m y + \Phi_m)$$

$$E_2(y,z,t) = E_0 \cos(\omega t - \beta_m z - \kappa_m y)$$

$$E(y,z,t) = E_1 + E_2 = 2E_0 \cos\left(\kappa_m y + \frac{1}{2}\Phi_m\right)\cos\left(\omega t - \beta_m z + \frac{1}{2}\Phi_m\right)$$

It is a stationary wave along the *y*-direction, which travels down the waveguide along *z*.

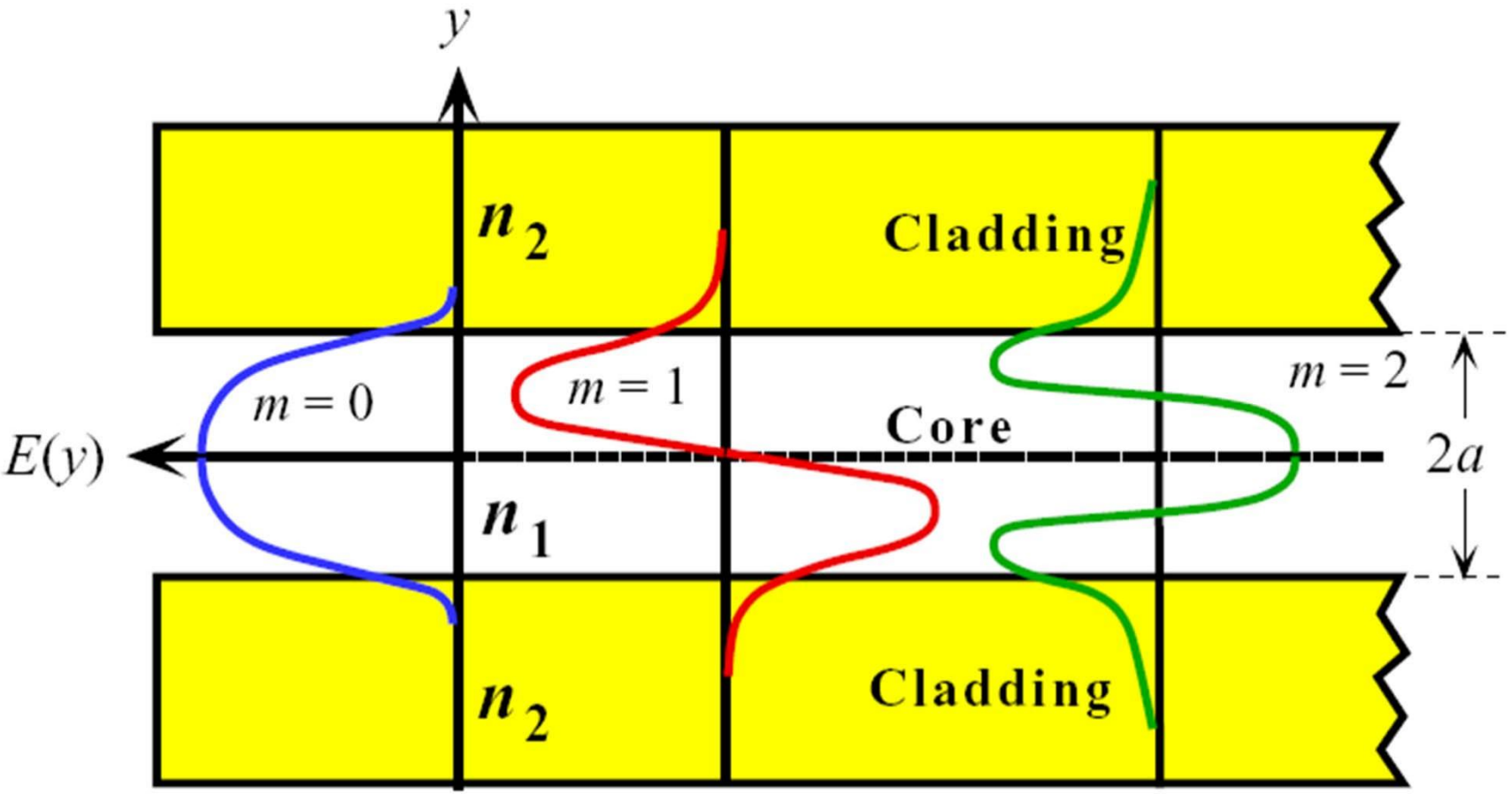
Electric Field Patterns



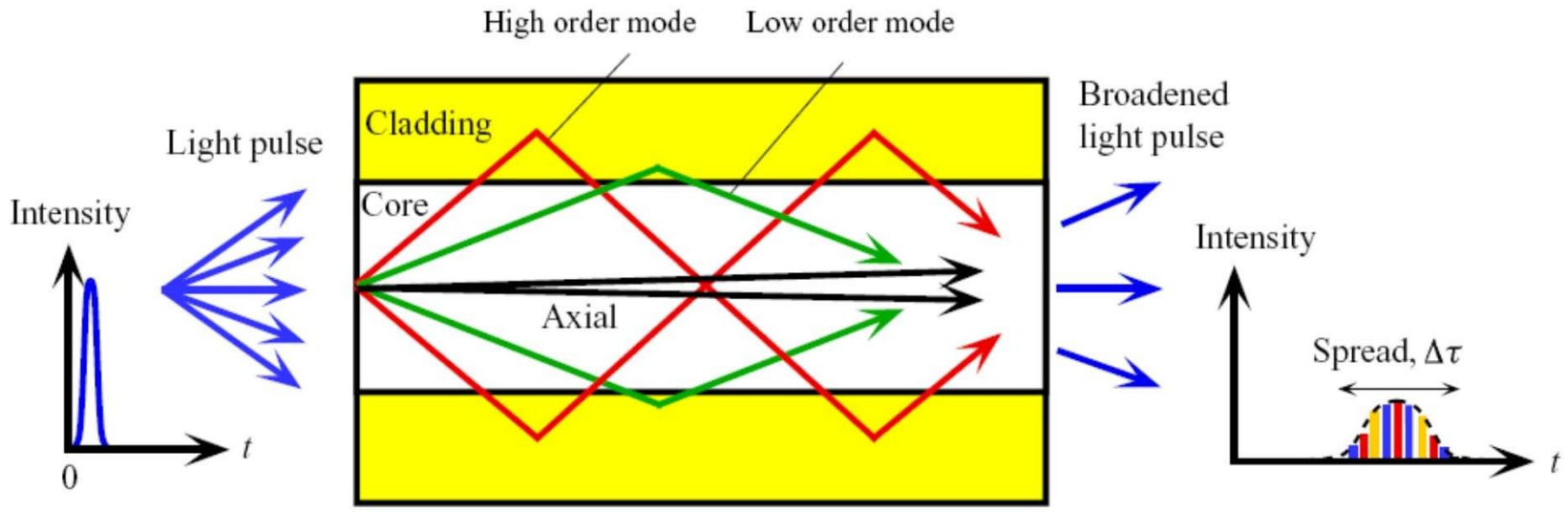
The lowest mode (m = 0) has a maximal intensity at the center and moves along z with a propagation constant of β_0. There is a propagating evanescent wave in the cladding near the boundary.

The two components, the electric field and the magnetic field form patterns across the fiber. These patterns are called **modes** of transmission. The **mode** of a fiber refers to the number of paths for the light rays within the cable

Electric Field Patterns



Broadening of the Output Pulse



Each light wave that satisfies the waveguide condition constitutes a **mode of propagation**. The integer *m* identifies these modes and is called the **mode number**.

θ_m is smaller for larger *m*. Higher modes exhibit more reflections and penetrate more into the cladding.

Short-duration pulses of light transmitted through the waveguide will be broadened in terms of time duration.

Single and Multimode Waveguides

$$\left[\frac{2\pi n_1(2a)}{\lambda}\right] cos\theta_m - \phi_m = m\pi$$

θ_m must nonetheless satisfy TIR,

$$\theta_m \geq \theta_c$$

$$sin\theta_m \geq \frac{n_2}{n_1}$$

$$\Rightarrow 1 - cos^2\theta_m \geq \frac{{n_2}^2}{{n_1}^2}$$

$$\Rightarrow cos^2\theta_m \leq 1 - \frac{{n_2}^2}{{n_1}^2}$$

$$\Rightarrow \frac{m\pi + \phi_m}{4\pi a/\lambda}$$

$$\leq ({n_1}^2 - {n_2}^2)^{1/2}$$

$$\Rightarrow \begin{cases} V = \dfrac{2\pi a}{\lambda}\left(n_1^2 - n_2^2\right)^{1/2} \\ m \leq (2V - \phi_m)/\pi \end{cases}$$

***V*-number (*V*-parameter, normalized thickness, normalized frequency)** is a characteristic parameter of the waveguide.

When $V < \pi/2$, $m = 0$ is the only possibility and only the fundamental mode ($m = 0$) propagates along the waveguide. It is termed as the **single-mode** waveguide. λ_c that satisfies $V = \pi/2$ is the **cut-off wavelength**. Above this wavelength, only the fundamental mode will propagate.

Example: waveguide modes

Consider a planar waveguide with a core thickness 20 μm, $n_1 = 1.455$, $n_2 = 1.440$, free-space light wavelength of 900 nm. Given the waveguide condition and the phase change in TIR for the TE mode, find angles θ_m for all the modes.

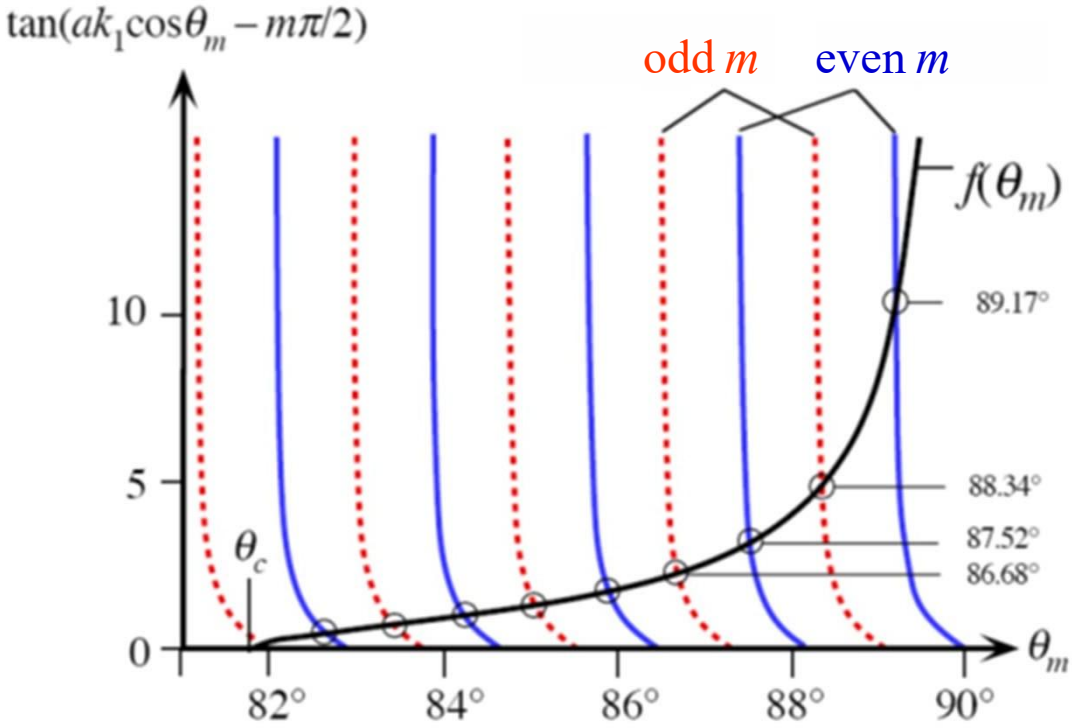
$$\left\lfloor \frac{2\pi n_1 (2a)}{\lambda} \right\rfloor \cos\theta_m - \phi_m = m\pi$$

$$\tan\left(\frac{1}{2}\phi_m\right) = \frac{\left\lfloor \sin^2\theta_m - \left(\frac{n_2}{n_1}\right)^2 \right\rfloor^{1/2}}{\cos\theta_m}$$

$$\tan\left(\frac{2\pi n_1 a}{\lambda}\cos\theta_m - m\frac{\pi}{2}\right) = \frac{\left\lfloor \sin^2\theta_m - \left(\frac{n_2}{n_1}\right)^2 \right\rfloor^{1/2}}{\cos\theta_m}$$

LHS (left-hand side): $\tan\left(\frac{2\pi n_1 a}{\lambda}\cos\theta_m - m\frac{\pi}{2}\right)$ *versus* θ_m

RHS (right-hand side): $f(\theta_m) = \frac{\left[\sin^2\theta_m - \left(\frac{n_2}{n_1}\right)^2\right]^{1/2}}{\cos\theta_m}$ *versus* θ_m



Penetration depth of the evanescent wave:

$$\delta_m = \frac{\lambda}{2\pi}\left(n_1^2 \sin^2\theta_m - n_2^2\right)^{-1/2}$$

m	0	1	2	3	4	5	6	7	8	9
θ_m	89.2°	88.3°	87.5°	86.7°	85.9°	85.0°	84.2°	83.4°	82.6°	81.9°
δ_m	0.691	0.702	0.722	0.751	0.793	0.866	0.970	1.15	1.57	3.83

δ_m in μm.

An accurate solution of the angle for the fundamental TE mode is 89.172°. The angle for the fundamental TM mode is 89.170°, which is almost identical to the angle for the TE mode.

Example: the number of modes

Estimate the number of modes that can be supported in a planar dielectric waveguide that is 100 μm wide and has $n_1 = 1.490$, $n_2 = 1.470$. The free-space light wavelength is 1 μm.

$$m \le \frac{2V - \phi}{\pi} \approx \frac{2V}{\pi}$$

$$V = \frac{2\pi a}{\lambda}\left(n_1^2 - n_2^2\right)^{1/2}$$

$$= \frac{100\pi}{1}\left(1.490^2 - 1.470^2\right)^{1/2} = 76.44$$

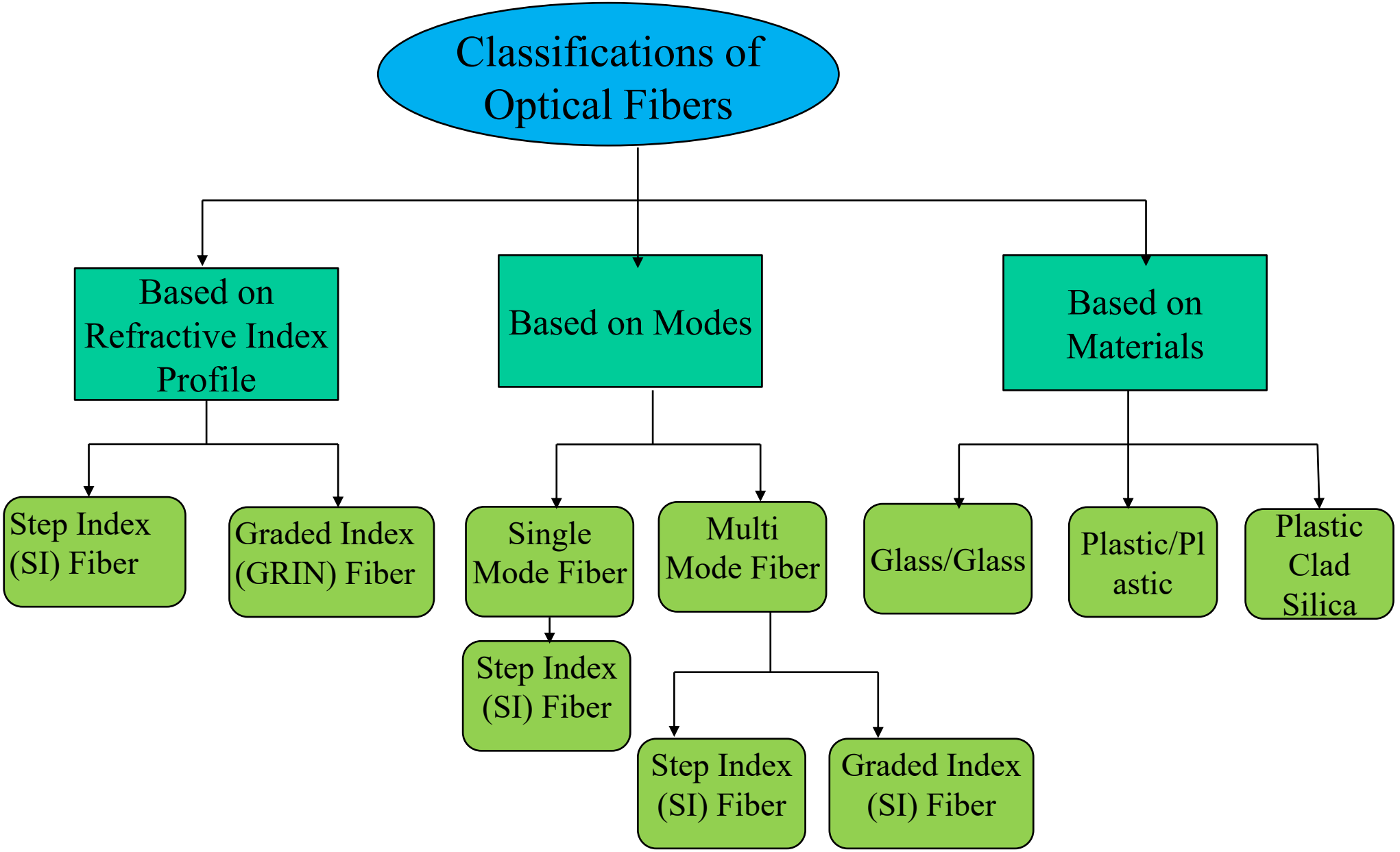
$$m \le \frac{2 \times 76.44}{\pi} = 48.7$$

There are about 49 modes.

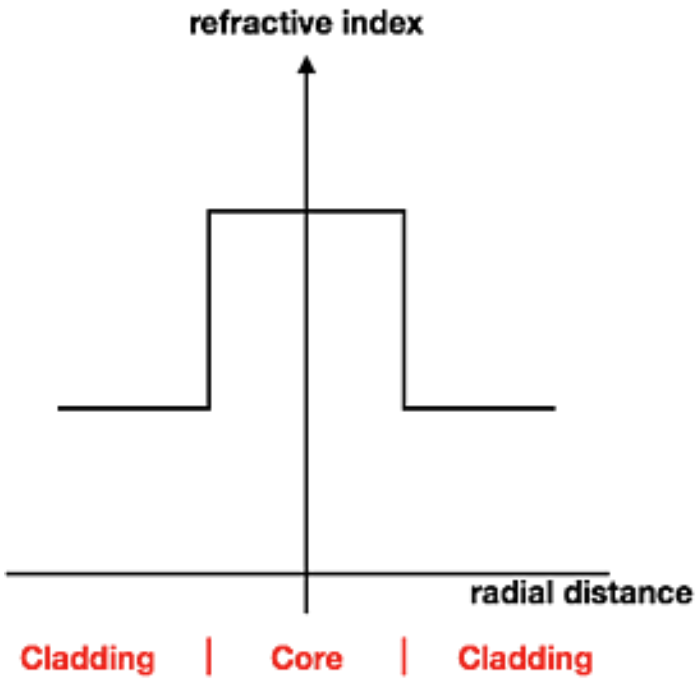
$$M = \text{Int}\left(\frac{2V}{\pi}\right) + 1$$

Int (x) is the integer function. It removes the decimal fraction of x.

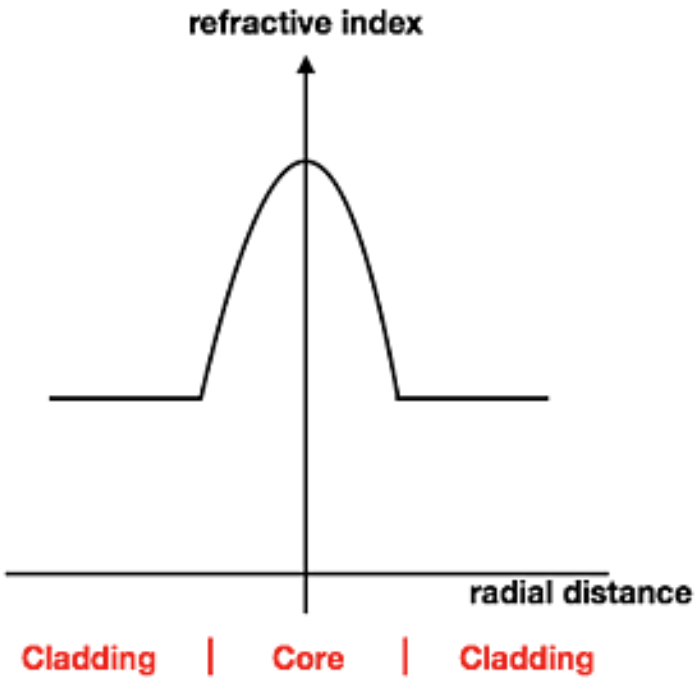
Classification of Fiber



Refractive Index Profile

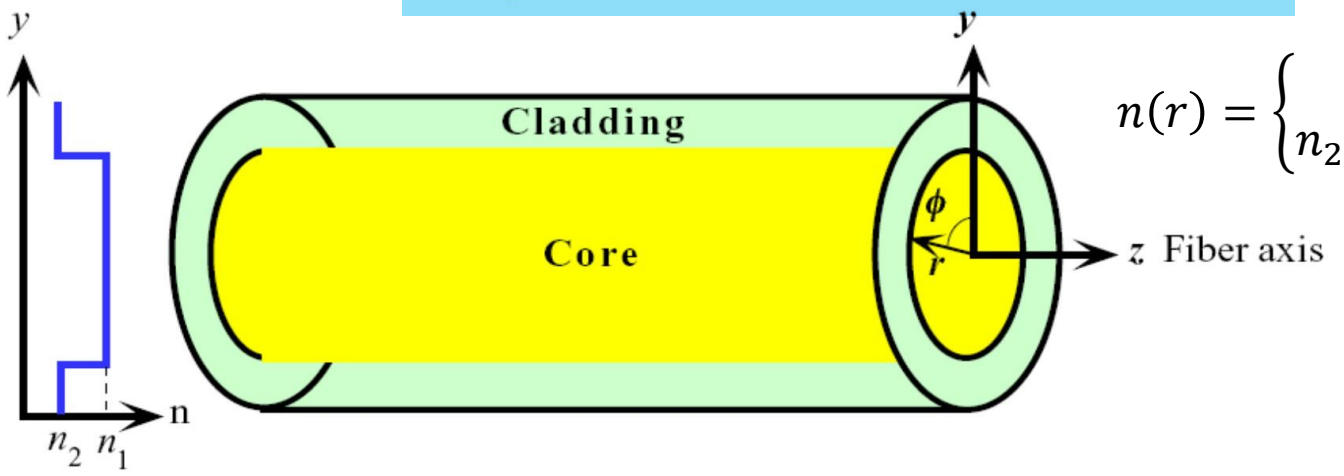


Step-Index



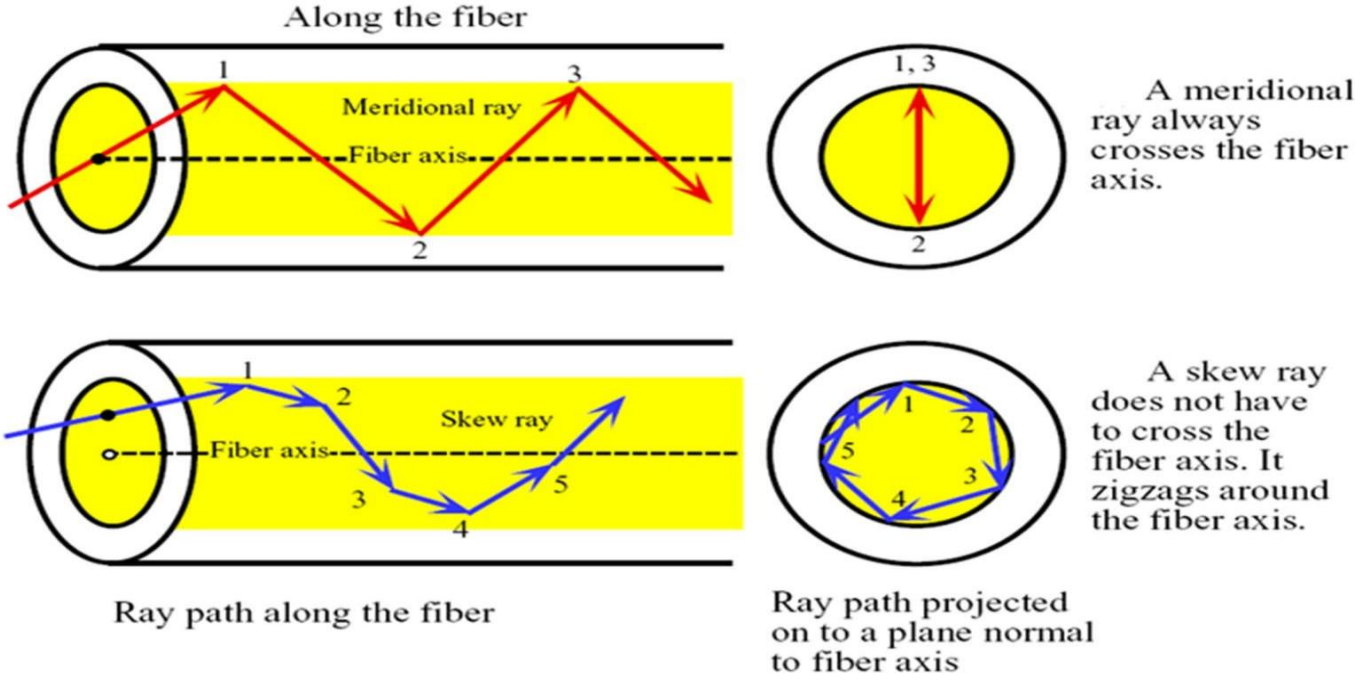
Graded-Index

Step-Index fiber

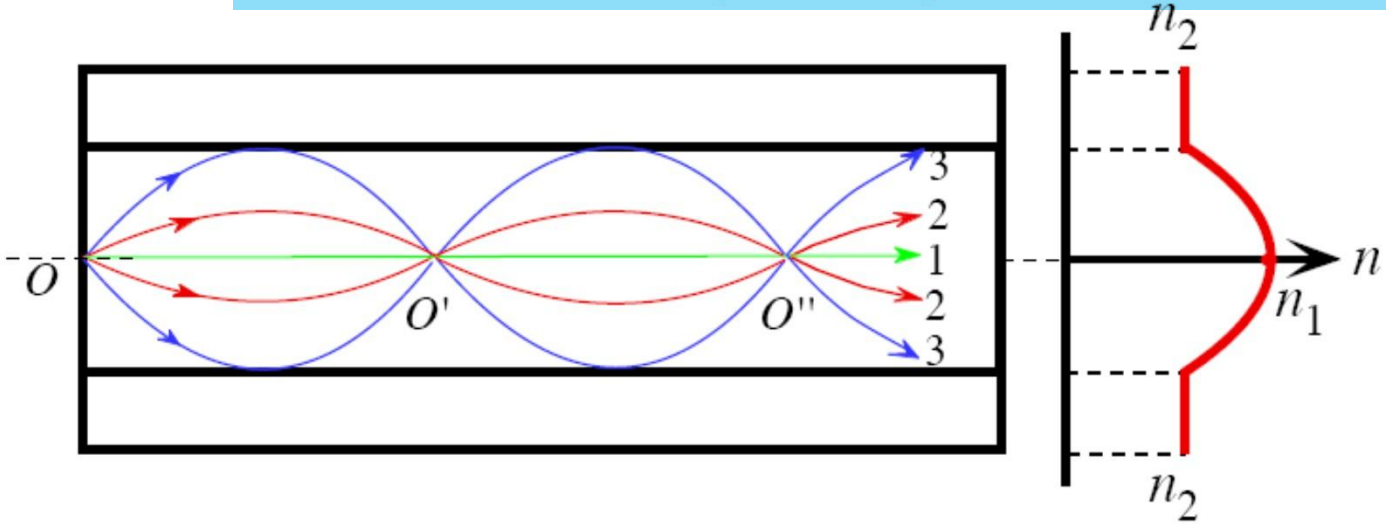


$$n(r) = \begin{cases} n_1 \ when\, r < a\ (core) \\ n_2 \ when\, r > a\ (cladding) \end{cases}$$

Normalized index difference, $\Delta = \frac{n_1 - n_2}{n_1}$ for practical fibers, $\Delta << 1$



Graded Index (GRIN) Fibers



a is the radius of the core,

r is the radial distance from the core axis

α shows characteristic of the refractive index profile

n_1 and n_2 are the refractive indices of core and cladding respectively.

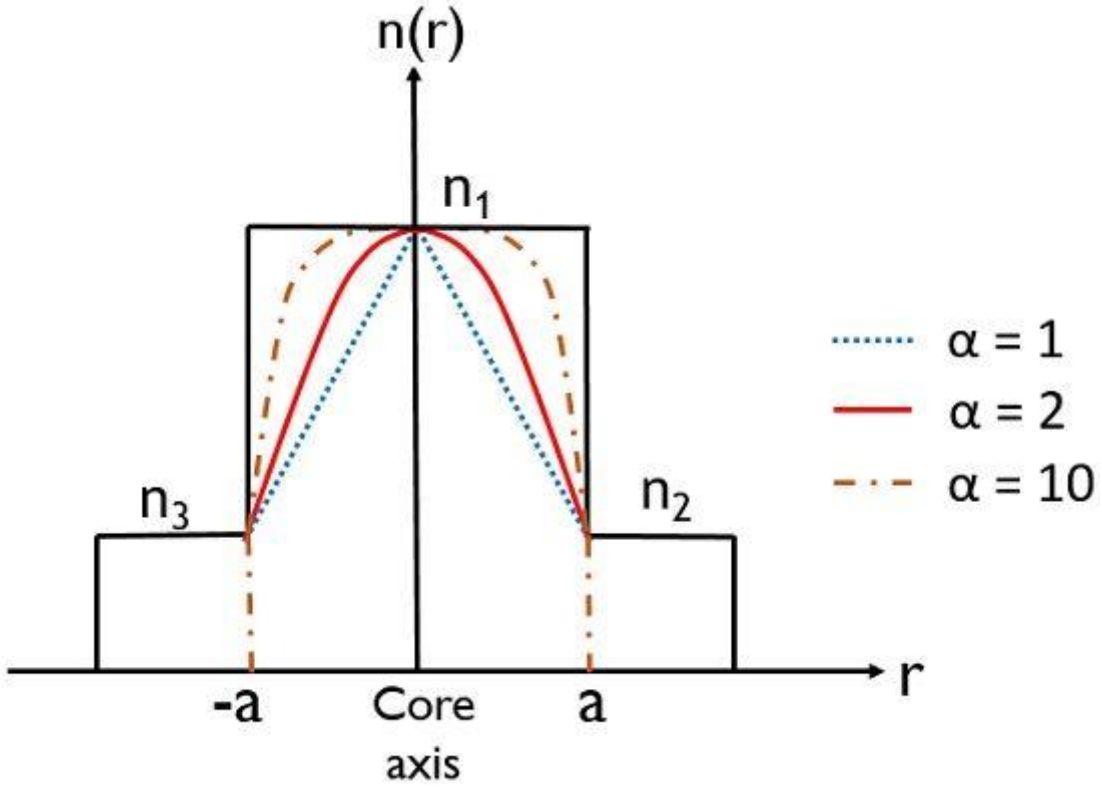
$$n(r) = \begin{cases} n_1 \left[1 - 2\Delta\left(\frac{r}{a}\right)^{\alpha}\right]^{1/2} & \text{for } 0 \le r \le a \\ n_1 (1 - 2\Delta)^{1/2} \approx n_1 (1 - \Delta) = n_2 & \text{for } r \ge a \end{cases}$$

In the **graded index (GRIN)** fiber, the refractive index is not constant within the core but decreases from n_1 at the center, as a power law, to n_2 at the cladding. Such a refractive index profile is capable of minimizing intermodal dispersion.

Intuitively, the velocity, c/n, is not constant and increases away from the center.

Graded Index (GRIN) Fibers

Profile parameter α determines the characteristic refractive index profile of fiber core. The range of refractive index as variation of α is shown in the following figure:



Single Mode Fiber

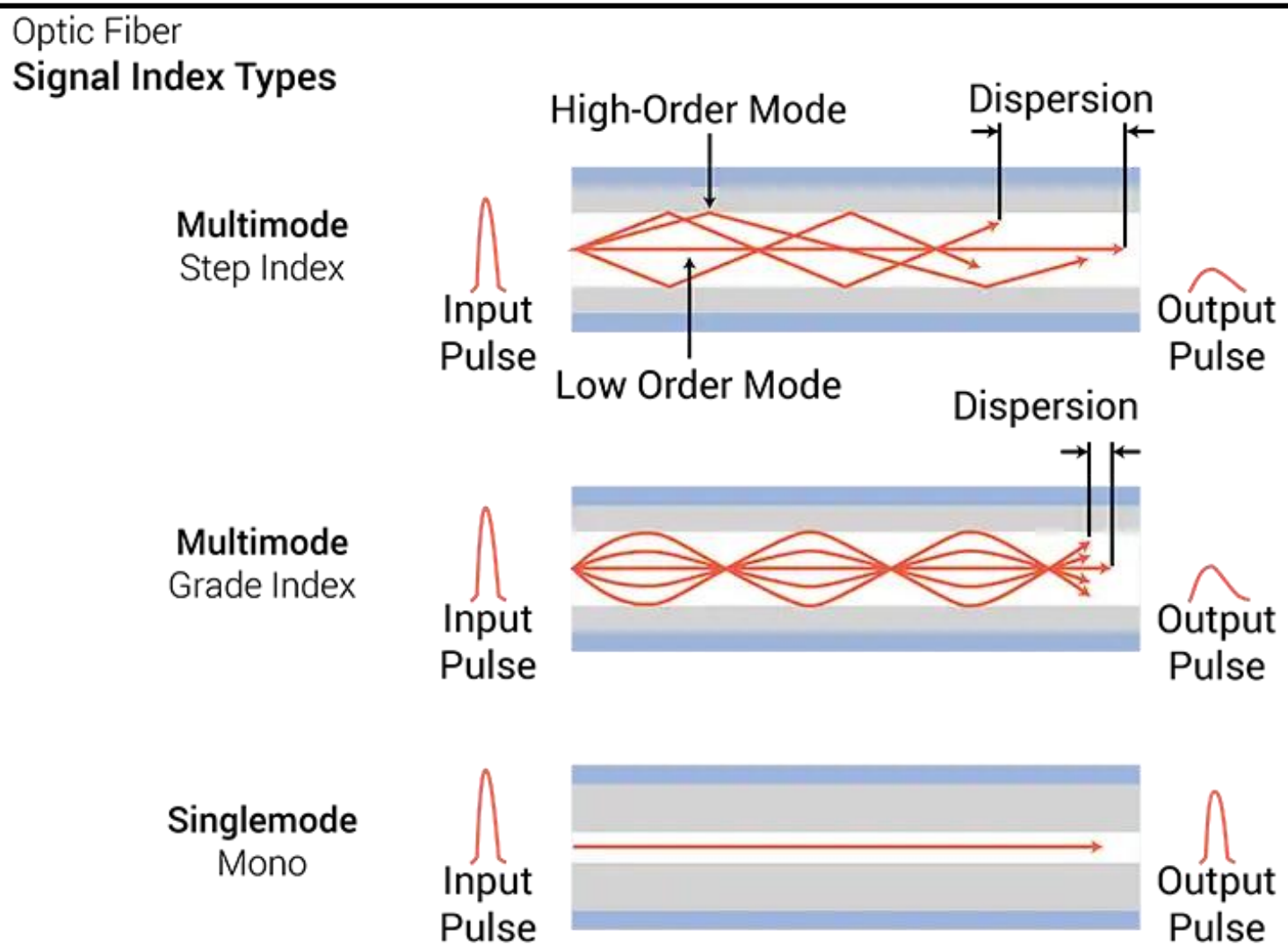
- Single mode fiber will permit only one mode to propagate and does not suffer from mode delay differences
- Typical core sizes are 2 to 15 µm. Single mode fiber is also known as fundamental or monomode fiber
- These are primarily developed for the 1300 nm window but they can be also be used effectively with time division multiplex (TDM) and wavelength division multiplex (WDM) systems operating in 1550 nm wavelength region



9/125 Single Mode

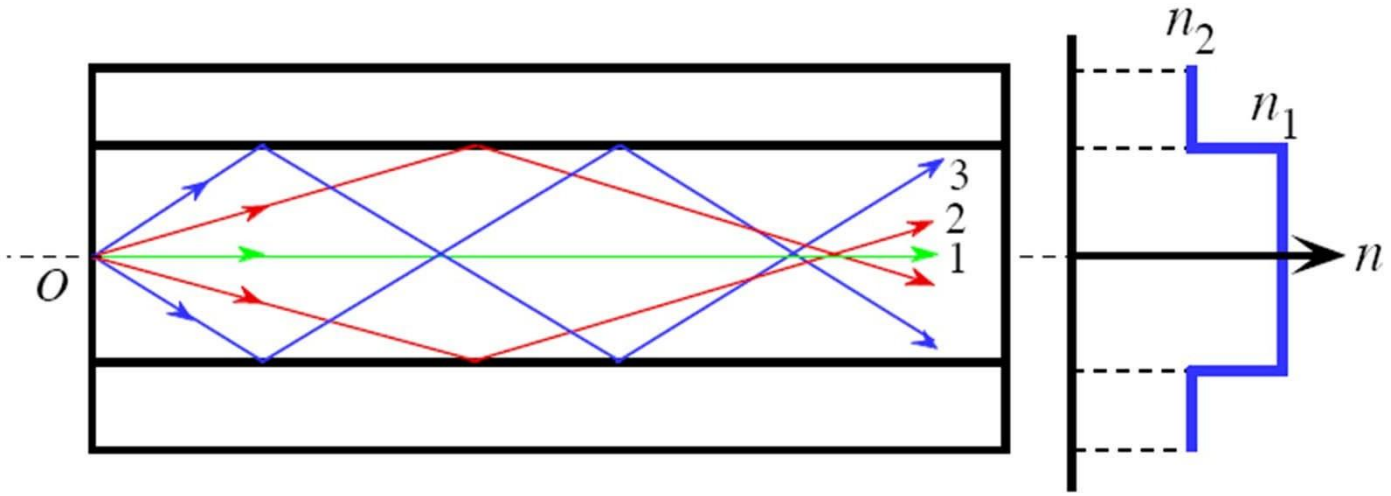
Multi-Mode Fiber

Multi-mode fiber is a type of optical fiber designed to carry multiple light rays or modes simultaneously, each at a marginally different reflection angle inside the optical fiber core.



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Drawback of Single Mode Fibers



Single mode step index fibers have small NAs and the amount of light coupled into a fiber is limited.

Multimode step index fibers have large NAs and core diameters, which allow for more light power launched into a fiber. However, they suffer from intermodal dispersion.

Intuitively, those rays that experience less reflections will arrive at the end of the fiber earlier.

Comparison between Single mode and Multi-mode Fiber

TABLE Comparison of typical characteristics of multimode step-index, single-mode step-index, and graded-index fibers. (Typical values combined from various sources.)

Property	Multimode step-index fiber	Single-mode step-index fiber	Graded index fiber
$\Delta = (n_1 - n_2)/n_1$	0.02	0.003	0.015
Core diameter (μm)	100	8.3 (MFD = 9.3 μm)	62.5
Cladding diameter (μm)	140	125	125
NA	0.3	0.1	0.26
Bandwidth × distance or Dispersion	20 − 100 MHz km.	<3.5 ps km^{-1} nm^{-1} at 1.3 μm >100 Gb s^{-1} km in common use	300 MHz km − 3 GHz km at 1.3 μm at 1.3 μm
Attenuation of light	4 − 6 dB km^{-1} at 850 nm 0.7 − 1 dB km^{-1} at 1.3 μm	1.8 dB km^{-1} at 850 nm 0.34 dB km^{-1} at 1.3 μm 0.2 dB km^{-1} at 1.55 μm	3 dB km^{-1} at 850 nm 0.6 − 1 dB km^{-1} at 1.3 μm 0.3 dB km^{-1} at 1.55 μm
Typical light source	Light emitting diode (LED)	Lasers, single mode injection lasers	LED, lasers
Typical applications	Short haul or subscriber local network communications	Long haul communications	Local and wide-area networks. Medium haul communications

V-Number of Step Index Fibers

$$V = \frac{2\pi a}{\lambda}\left(n_1^2 - n_2^2\right)^{1/2}$$

a is the radius of the core.

$$\Delta = (n_1 - n_2)/n_1 \qquad n_1 > n_2$$

$$n = (n_1 + n_2)/2 \qquad \Delta << 1$$

$$V = \frac{2\pi a}{\lambda}\left[(n_1 + n_2)(n_1 - n_2)\right]^{1/2} = \frac{2\pi a}{\lambda}\left(2n_1 n\Delta\right)^{1/2}$$

When the *V*-number is smaller than **2.405**, only the fundamental mode (LP_{01}) can propagate through the fiber core (**single-mode fiber**). The cut-off wavelength λ_c above which the fiber becomes single-mode is given by

$$V_{\text{cut-off}} = \frac{2\pi a}{\lambda_c}\left(n_1^2 - n_2^2\right)^{1/2} = 2.405$$

The number of modes M in a step index fiber can be estimated by

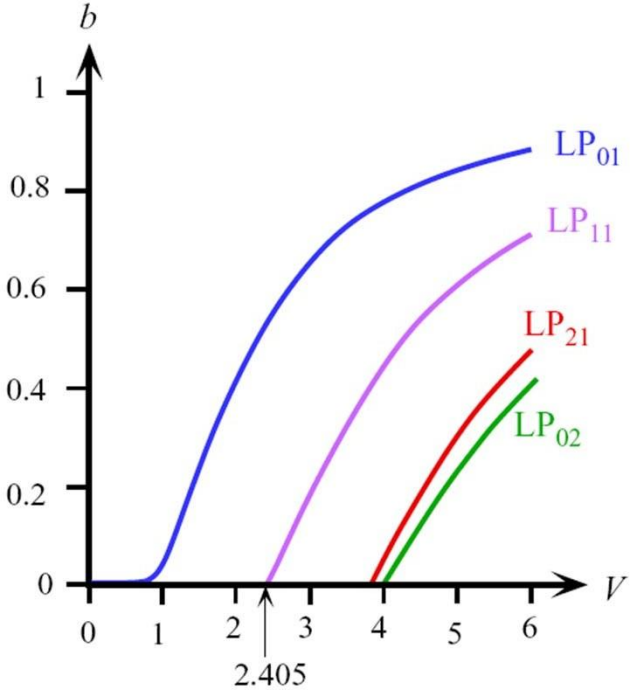
$$M \approx \frac{V^2}{2}$$

Since the propagation constant β_{lm} depends on the waveguide properties and wavelength, a **normalized propagation constant** is usually defined. This normalized propagation constant, b, depends only on the V-number.

$$b = \frac{\frac{\lambda\beta_{lm}}{2\pi} - {n_2}^2}{{n_1}^2 - {n_2}^2}$$

b = 0 corresponds to $\beta_{lm} = 2\pi n_2/\lambda$

b = 1 corresponds to $\beta_{lm} = 2\pi n_1/\lambda$



Example: A Multimode Fiber

A step-index fiber has a core of refractive index 1.468 and diameter 100 μm, a cladding of refractive index of 1.447. If the source wavelength is 850 nm, calculate the number of modes that are allowed in this fiber.

$$V = \frac{2\pi a}{\lambda}\sqrt{n_1^2 - n_2^2} = \frac{\pi 100}{0.85}\sqrt{1.468^2 - 1.447^2} = 91.44$$

$$M \approx \frac{V^2}{2} = \frac{91.44^2}{2} = 4181$$

Example: A Single Mode Fiber

What should be the core radius of a single mode fiber that has a core refractive index of 1.468 and a cladding refractive index of 1.447, and is to be used for a source wavelength of 1.3 μm?

$$V = \frac{2\pi a}{\lambda}\sqrt{{n_1}^2 - {n_2}^2} = \frac{2\pi a}{1.3}\sqrt{1.468^2 - 1.447^2} \leq 2.405$$

$$a \leq 2.01\,\mu\text{m}$$

Example: Single Mode Cut-Off Wavelength

What is the cut-off wavelength for single mode operation for a fiber that has a core with a diameter of 7 μm, a refractive index of 1.458, and a cladding of refractive index of 1.452? What is the V-number and the mode field diameter (MFD) when operating at $\lambda = 1.3$ μm?

$$V = \frac{2\pi a}{\lambda}\sqrt{n_1^2 - n_2^2} = \frac{\pi 7}{\lambda}\sqrt{1.458^2 - 1.452^2} \leq 2.405$$

$$\lambda \geq 1.208\,\mu\text{m}$$

When $\lambda = 1.3$ μm:

$$V = \frac{2\pi a}{\lambda}\sqrt{n_1^2 - n_2^2} = \frac{\pi 7}{1.3}\sqrt{1.458^2 - 1.452^2} = 2.235$$

$$2w_0 \approx (2a)\frac{V+1}{V} = 7 \times \frac{2.235+1}{2.235} = 10.13\,\mu\text{m}$$